9A Lesson 1

Environmental variation

Year 9 Science · 9A Genetics and evolution · Student Book pages 5 to 7

Name _____ Class _____ Date _____

Today I can:

- ☐ I can describe abiotic and biotic environmental factors and give examples of environmental variation.
- ☐ I can tell continuous variation from discontinuous variation, and explain why environmental variation can make classification difficult.

Key words 关键词

species	物种
extinct	灭绝的
environment	环境
abiotic factor	非生物因素
biotic factor	生物因素
characteristic	特征
variation	变异
environmental variation	环境变异
continuous variation	连续变异
discontinuous variation	不连续变异
classification	分类

Command words 指令词

Complete 完成（填写） · Describe 描述 · Draw 画出 · Explain 解释 · Give 给出 · Give a reason 给出理由 · Label 标注 · Match 连线（配对） · Plot 绘制（数据点） · Suggest 提出（建议） · Tick 打勾 · Use 使用（运用）

● Core Everyone does this section.

1. Match each key word to its meaning. Draw a line.

species	the surroundings of an organism
extinct	a non-living factor, e.g. temperature or light
environment	a species with no living members left
abiotic factor	a group of organisms that can breed together to produce offspring that can also breed
biotic factor	the activities of other organisms, e.g. predation or disease

2. Tick one box in each row to show whether the factor is abiotic or biotic.

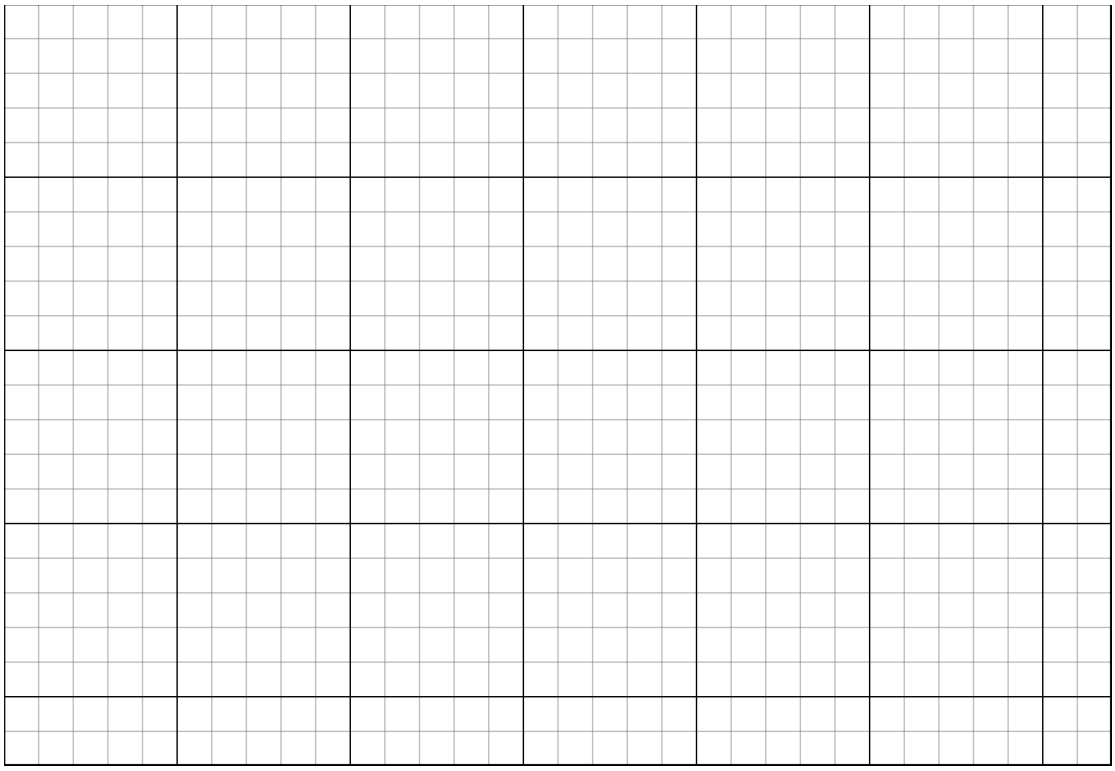
Environmental factor	Abiotic	Biotic
temperature		
predation (being hunted)		
amount of light		
competition for food		
rainfall		
infectious disease		

3. Twenty potatoes of the same variety were weighed. Their masses in grams were: 142, 118, 165, 151, 137, 174, 129, 158, 146, 183, 121, 160, 149, 133, 171, 155, 140, 127, 163, 148.

a) Complete the tally chart.

Mass range (g)	Tally	Number of potatoes
110 to 129		
130 to 149		
150 to 169		
170 to 189		

b) Plot a bar chart of the number of potatoes in each mass range on the grid below. Label both axes.



c) Is potato mass continuous or discontinuous variation? Give a reason.

●● Stretch When Core is finished.

S1. Describe two examples of environmental variation in humans and state the environmental factor that causes each one.

Example of environmental variation	Environmental factor that causes it

S2. Tick one box in each row to show whether the variation is continuous or discontinuous.

Characteristic	Continuous	Discontinuous
height		
blood group		
hand span		
number of fingers		
length of a leaf		
able to roll the tongue (yes or no)		

S3. The twenty potatoes in question 3 were all the same variety. Explain why they still varied in mass.

●●● Challenge Explain in full sentences.

C1. A plant on a seashore has pink flowers and is 10 cm tall. On a nearby island, similar plants are over 50 cm tall with dark purple flowers. Scientists grew seeds from both places in the same laboratory. All the plants grew to look the same.

a) What does this tell you about the plants from the two areas?

b) Use this example to explain how environmental variation can make it difficult to identify plant species.

C2. In 2003, bones of adult humans just over 1 m tall were found on Flores Island. Some scientists think they were our species but had too little iodine in their diet; others think they were a different species that is now extinct. Suggest what evidence would help scientists decide, and explain your reasoning.

My thinking (any language 可用中文)

A classmate says: 'All variation is caused by the environment.' Do you agree? Explain why, or give an example that shows they are wrong.

Word bank abiotic factor 非生物因素 · biotic factor 生物因素 · environment 环境 · environmental variation 环境

变异 · continuous variation 连续变异 · discontinuous variation 不连续变异 · species 物种 · classification 分类

✂ ---------- cut here

Exit ticket 9A Lesson 1: Environmental variation Name _____

1. Give one abiotic factor and one biotic factor in a pond.

_____

2. Is a scar an example of inherited or environmental variation?

_____

3. Is foot length continuous or discontinuous variation? Give a reason.

_____

Today I can: Goal 1 ☐ Goal 2 ☐